Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Bo Zhu	**Email**	bo.zhu@gatech.edu
Institute Name	Georgia Institute of Technology			
Address	756 W Peachtree St NW, Atlanta, GA 30308			

2. Technology Overview

Title of Proposed Technology	***A Physical AI Infrastructure for Generative Digital Twin Training and Deployment***				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ),** **Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√),** **Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Information & Communication			***Mechanical · Materials***	
	Medium Category		Small Category	Medium Category	Small Category
	Digital Content		***Computer Graphics***	Robotics / Automation Machinery	Robotics/Automation Machinery Software
Concept of Proposed Technology	*The proposed technology aims to develop a* ***Physical AI infrastructure for the automatic generation, training, and deployment of generative digital twins in industrial systems, grounded in first-principles physical simulation and physics-embedded machine learning****. The core objective of this infrastructure is to substantially reduce reliance on manual simulation programming and model-specific engineering by unifying high-fidelity physical simulation, structure-preserving generative modeling, and automated computational pipelines. Rather than treating digital twin development as a sequence of ad hoc, case-specific modeling decisions, the proposed approach establishes a principled infrastructure that supports scalable, repeatable, and systematic digital twin construction across diverse industrial*				

(max 10 pages)

	scenarios. *The proposed infrastructure is built around three tightly coupled capabilities that reflect a physics-based approach to generative modeling of complex systems. (1)* ***Automated generative training of complex physical systems****, where training objectives and supervision signals are derived directly from physically grounded simulations and geometric system representations. This design leverages mature simulation methodologies to generate informative training data, significantly reducing human intervention while maintaining physical fidelity. (2)* ***Physically consistent state evolution modeling****, which focuses on learning long-horizon system dynamics through generative models that explicitly preserve physical structure, constraints, and stability properties. By modeling transport, interaction, and evolution at the level of physically meaningful state representations, this capability enables reliable prediction beyond short-term rollouts where conventional surrogate models typically degrade. (3)* ***Efficient adaptation and updating mechanisms****, which enable digital twins to incorporate new observations and evolving operating conditions through localized or incremental updates, avoiding costly full retraining and supporting continuous refinement in dynamic industrial environments.* *In contrast to conventional digital twin pipelines that rely on tightly coupled, manually engineered simulation workflows or purely data-driven surrogates, the proposed Physical AI infrastructure treats digital twin training as an automated, physically constrained generative process. By embedding physical priors, geometric structure, and stability considerations directly into the generative modeling pipeline, the infrastructure bridges the gap between high-fidelity simulation and data-driven learning. This allows digital twins to remain stable, interpretable, and robust under partial observability and limited data regimes that commonly arise in practice. As a result, the proposed infrastructure provides a practical and scalable foundation for deploying generative digital twins in real-world industrial systems, supporting monitoring, prediction, and decision-making tasks while maintaining physical consistency, robustness, and computational efficiency.*
Rationale for Joint R&D Project	*The proposed project requires close and sustained collaboration between a research institute and a Korean SME partner to successfully develop and deploy a Physical AI infrastructure for generative digital twin training in industrial systems. The research institute brings deep expertise in high-fidelity and high-performance physical simulation, physics-embedded generative modeling, and long-horizon state evolution, which are essential for defining the core algorithms and representations underlying the proposed infrastructure. However, the transformation of these research capabilities into an infrastructure that is usable, robust, and commercially viable for industrial applications fundamentally depends on the active involvement of an industrial partner with direct ownership of real systems, workflows, and deployment constraints.* *The necessity of a joint R&D framework arises from three tightly coupled requirements that cannot be addressed in isolation. (1) Industrial grounding of physical models and representations, where the Korean SME partner defines concrete industrial application scenarios, provides realistic system configurations, and specifies operational constraints that determine which physical effects, scales, and*

(max 10 pages)

	performance trade-offs must be captured by the digital twins. These requirements are highly domain-specific and cannot be inferred from academic benchmarks or simplified simulation settings. (2) Bridging physically grounded research methods to deployable infrastructure, where the research institute contributes foundational advances in physically consistent generative modeling and long-horizon simulation, while the Korean SME partner ensures that these methods are integrated into existing industrial simulation pipelines, software environments, and computing infrastructures. This bidirectional interaction is essential for preserving physical fidelity while meeting practical constraints on efficiency, robustness, and integration. (3) Iterative co-design and validation in realistic operating regimes, in which continuous feedback between algorithm development and industrial evaluation—led by the Korean SME partner—drives progressive refinement of the infrastructure toward reliability, scalability, and maintainability under real-world operating conditions. *Such tight co-design between Physical AI methodologies and industrial deployment pipelines is critical for addressing system-level challenges including numerical stability, computational efficiency, data availability, and integration compatibility from the earliest stages of development. These challenges cannot be effectively resolved through sequential technology transfer after research completion, and instead require sustained joint development in which industrial constraints directly inform research priorities and design choices.*
Key Words	*1. Physical AI Infrastructure* *2. Generative Digital Twin* *3. High-Performance Physical Simulation* *4. Generative Modeling of Complex Physical Systems* *5. Industrial Digital Twin Systems*

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		**Start of the Project**	Level (1-2)	**End of the Project**	Level (6)
R&D Objectives		*The objective of this R&D project is to develop a Physical AI infrastructure that enables reliable, scalable, and automated generative digital twin training and deployment for industrial systems, by tightly integrating high-performance physical simulation with generative modeling under explicit physical constraints. The project aims to substantially reduce manual simulation programming and model-specific engineering while ensuring physical consistency, numerical stability, and practical applicability in industrial environments.* *To achieve this goal, the project is organized around three interrelated research objectives. (1)* ***Long-horizon generative modeling of complex physical systems****, focusing on the development of generative modeling methodologies that learn physically meaningful state evolution over long time horizons while preserving essential physical properties such as conservation laws, stability, and structural invariants. This objective targets the fundamental challenge of enabling accurate and robust digital twins beyond short-term prediction regimes. (2)* ***Physics-embedded training and verification mechanisms****, which establish infrastructure-level methods for incorporating physical constraints, stability criteria, and system-specific requirements directly into the training and inference process. This objective ensures that generative digital twins exhibit trustworthy behavior under limited data, partial observability, and varying operating conditions common in industrial settings. (3)* ***Deployment-oriented infrastructure and automation****, which focuses on building an integrated infrastructure that supports automated digital twin training, efficient updating, and seamless integration with existing industrial simulation and monitoring workflows. This objective enables scalable deployment and continuous refinement of generative digital twins without costly full retraining.*			
R&D Target Specifications		*The performance targets are organized along four key dimensions.* *(1)* ***Physical modeling accuracy and consistency****, where the generative digital twins are expected to achieve predictive errors within 5–10% relative to high-fidelity reference simulations for representative industrial systems, while preserving essential physical invariants such as conservation laws and stability over long-horizon evolution.* *(2)* ***Computational efficiency and scalability****, where the infrastructure is expected to deliver performance improvement in simulation throughput compared to conventional high-fidelity pipelines, enabling near–real-time or real-time deployment depending on system complexity without sacrificing physical consistency.* *(3)* ***Automation and reduction of manual engineering effort****, where the project targets a significant reduction of in hand-written, system-specific simulation code relative to baseline industrial digital twin workflows, reflecting the effectiveness of automated training and configuration generation.*			

(max 10 pages)

		*(4) **Robustness under realistic operating conditions**, where trained digital twins are expected to maintain stable performance under varying configurations, partial observations, and moderate parameter perturbations typical of industrial environments, without requiring costly full retraining.*
R&D Plan		*This R&D project will be conducted over a period of three years, following a progressive development and validation strategy that advances from simulation-based development to industrially relevant integration and finally to deployment-oriented validation in real-world industrial systems. The plan is designed to incrementally increase system complexity, realism, and deployment fidelity while continuously validating physical consistency, robustness, and computational efficiency.* *During **Year 1**, the focus will be on simulation-based development and core Physical AI infrastructure construction. This phase will establish an integrated high-performance physical simulation environment that serves as the foundation for automated generative digital twin training. Core generative modeling components and physically grounded state evolution representations will be developed and validated in controlled simulation settings. Emphasis will be placed on establishing physically constrained training pipelines, validating long-horizon stability, and benchmarking predictive accuracy and computational performance against high-fidelity reference simulations.* *During **Year 2**, the project will advance to laboratory-scale and industrial workflow integration. The Physical AI infrastructure will be integrated with representative industrial simulation pipelines and system configurations provided by the Korean SME partner. This phase will focus on adapting the infrastructure to practical computational constraints, data availability conditions, and system variability encountered in industrial environments. Iterative validation will be performed to evaluate robustness, automation effectiveness, and reliability of generative digital twin training and updating under realistic operating scenarios.* *During **Year 3**, the project will concentrate on industrial deployment and technology maturation. The fully integrated Physical AI infrastructure will be deployed in industrially relevant settings to support generative digital twin training, updating, and inference for real-world systems. Large-scale evaluation will be conducted to assess predictive accuracy, stability under changing conditions, and deployment efficiency. Based on these evaluations, the infrastructure will be further refined to improve usability, performance, and integration readiness. This phase will also assess commercialization potential and prepare the technology for industrial transfer, with a focus on sustainable deployment and scalability within Korean industrial ecosystems.*

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The research institute, through the Computer Graphics Group at Georgia Tech, will be responsible for the development of the core Physical AI methodologies and infrastructure-level intelligence for the proposed generative digital twin framework. The group will lead research on high-performance physical simulation, physics-embedded generative modeling, and long-horizon state evolution, which collectively form the scientific foundation of the proposed infrastructure.* *Specifically, the Computer Graphics Group will focus on: (1) the development of physics-grounded generative models for complex physical systems, incorporating physical constraints and stability properties; (2) tight coupling of generative models with high-fidelity physical simulation backbones for training, verification, and benchmarking; and (3) the design and automation of the Physical AI infrastructure, including automated digital training pipelines and software-level integration in simulation environments.* *Throughout the project, the research institute will provide technical leadership and collaborate closely with the Korean SME partner to support system integration, industrial adaptation, and experimental evaluation, ensuring effective transfer of research outcomes into an industrially deployable infrastructure.*
Expected R&R of Korean SME Partner	*The Korean SME partner will be responsible for the industrial implementation, validation, and benchmarking of the proposed Physical AI infrastructure for generative digital twin training and deployment. In particular, the SME partner will take the lead on: (1) providing or configuring industrial system setups, including access to existing industrial systems and data streams required for digital twin construction and validation; (2) integration of the Physical AI infrastructure with existing industrial simulation and monitoring pipelines, enabling large-scale testing and evaluation under realistic computational and operational constraints; (3) execution of laboratory-scale and pilot-scale testing, including validation of digital twin accuracy, robustness, and update behavior across different operating conditions and system configurations; and (4) rigorous benchmarking against predefined success criteria, ensuring that the developed infrastructure meets performance, reliability, and scalability requirements necessary for industrial deployment.*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*(1) Technical outcomes, where the project will produce an integrated infrastructure that tightly couples high-performance physical simulation with physically grounded generative modeling. The resulting system will demonstrate reliable long-horizon prediction, stable behavior under varying operating conditions, and substantial improvements in automation and computational efficiency relative to conventional digital twin pipelines. Successful validation in industrially relevant environments will establish the infrastructure's readiness for deployment-oriented use.* *(2) Industrial and commercial impact, where the project will provide the Korean SME partner with core Physical AI technologies, system integration expertise, and validated deployment workflows applicable to a broad class of industrial digital twin and simulation-driven applications. Through joint development and validation, the SME partner will strengthen its capability to design, deploy, and maintain next-generation digital twin solutions that significantly reduce manual engineering effort while preserving physical fidelity, stability, and reliability. These outcomes directly support commercialization and product development, enabling scalable and cost-effective adoption of generative digital twin technologies in industrial settings.* *(3) Broader ecosystem and national impact, where the project is expected to enhance the technological competitiveness of Korean SMEs in industrial simulation, digital twin infrastructure, and Physical AI–driven engineering. By establishing a reusable and trustworthy Physical AI infrastructure through international joint R&D, the project contributes to the advancement of Korea's industrial AI ecosystem and supports the broader adoption of high-performance, physically consistent AI technologies across multiple industrial domains.*

6. References

(1) Gao, Y., Yu, H.-X., Zhu, B., and Wu, J. (2025). FluidNexus: 3D fluid reconstruction and prediction from a single video. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).

(2) Li, Y., Sun, Y., Ma, P., Sifakis, E., Du, T., Zhu, B., and Matusik, W. (2024). Neural fluidic system design and control with differentiable simulation. Advances in Neural Information Processing Systems (NeurIPS).

(3) Yu, H.-X., Zheng, Y., Gao, Y., Deng, Y., Zhu, B., and Wu, J. (2024). Inferring hybrid neural fluid fields from videos. Advances in Neural Information Processing Systems (NeurIPS), 36.

(4) Deng, Y., Yu, H.-X., Zhang, D., Wu, J., and Zhu, B. (2023). Fluid simulation on neural flow maps. ACM Transactions on Graphics (TOG), 42(6), 1–21.

(5) Deng, Y., Yu, H.-X., and Wu, J., and Zhu, B. (2023). Learning vortex dynamics for fluid inference and prediction. International Conference on Learning Representations (ICLR).

(6) Xian, Z., Zhu, B., Xu, Z., Tung, H.-Y., Torralba, A., Fragkiadaki, K., and Gan, C. (2023). FluidLab: A differentiable environment for benchmarking complex fluid manipulation. International Conference on Learning Representations (ICLR).

(7) Deng, Y., Zhang, Y., He, X., Yang, S., Tong, Y., Zhang, M., DiPietro, D., and Zhu, B. (2021). Soft multicopter control using neural dynamics identification. Conference on Robot Learning (CoRL), 1773–1782.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)